

EXAM 7 – SPRING 2013

12. (2.5 points)

Beginning in 2013, an insurance company entered a niche market that other companies had not yet entered. The company plans to grow aggressively in this market. Given the following information for this insurance company:

Calendar Year Projections (\$000,000)

	<u>2013</u>	<u>2014</u>	<u>2015</u>
Beginning US GAAP equity	1,000	1,105	1,221
Net income	150	166	183
Dividends to be paid at year-end	45	50	55

- Risk-free rate: 2.0%
- Expected equity market risk premium: 6.0%
- Insurance company equity beta: 1.5
- CAPM is an appropriate model for determining the company's risk-adjusted discount rate.
- The company's plan is to continue its current dividend payout ratio.

a. (2 points)

Determine the value of this company as of January 1, 2013 based on the abnormal earnings valuation method. Assume that after the forecast horizon abnormal earnings will decrease to zero, linearly, by 2020.

b. (0.5 point)

Explain why an investor might prefer the abnormal earnings valuation method over the dividend discount model valuation for this particular company.

CONTINUED ON NEXT PAGE

Question 12 Sample Answer

Solution 1

- a) Required Return = $k = r_f + \beta(E[r_m] - r_f) = 2.0\% + 1.50(6.0\%) = 11.0\%$
 Abnormal Earnings (AE)₂₀₁₃ = $NI_{2013} - k \times BV_{1/1/2013} = 150.0 - 11.0\% \times 1,000 = 40.0$

$$AE_{2014} = NI_{2014} - k \times BV_{1/1/2014} = 166.0 - 11.0\% \times 1,105 = 44.5$$

$$AE_{2015} = NI_{2015} - k \times BV_{1/1/2015} = 183.0 - 11.0\% \times 1,221 = 48.7$$

$$\text{Total Equity Value}_{1/1/2013} = BV_{1/1/2013} + \Sigma PV(AE's)$$

Year	Abnormal Earnings		PV Factor = $1 / (1 + k)^{(\text{Year} - 2012)}$	PV of AE's
	Forecast Horizon	Beyond Forecast Horizon		
2013	40.0		90.1% = $1 / 1.11$	36.0
2014	44.5		81.2% = $1 / 1.11^2$	36.1
2015	48.7		73.1% = $1 / 1.11^3$	35.6
2016		39.0 = $48.7 \times 4 / 5$	65.9% = $1 / 1.11^4$	25.7
2017		29.2 = $48.7 \times 3 / 5$	59.3% = $1 / 1.11^5$	17.3
2018		19.5 = $48.7 \times 2 / 5$	53.5% = $1 / 1.11^6$	10.4
2019		9.7 = $48.7 \times 1 / 5$	48.2% = $1 / 1.11^7$	4.7
			BV _{1/1/2013}	1,000.0
Total Value				1,165.8

Solution 2

- a) Required Return = $k = r_f + \beta(E[r_m] - r_f) = 2.0\% + 1.50(6.0\%) = 11.0\%$

$$\text{Abnormal Earnings (AE)}_{2013} = NI_{2013} - k \times BV_{1/1/2013} = 150.0 - 11.0\% \times 1,000 = 40.0$$

$$AE_{2014} = NI_{2014} - k \times BV_{1/1/2014} = 166.0 - 11.0\% \times 1,105 = 44.5$$

$$AE_{2015} = NI_{2015} - k \times BV_{1/1/2015} = 183.0 - 11.0\% \times 1,221 = 48.7$$

$$\text{Total Equity Value}_{1/1/2013} = \text{BV}_{1/1/2013} + \Sigma \text{PV}(\text{AE's})$$

Year	Abnormal Earnings		PV Factor = $1 / (1 + k)^{(\text{Year} - 2012)}$	PV of AE's
	Forecast Horizon	Beyond Forecast Horizon		
2013	40.0		90.1% = $1 / 1.11$	36.0
2014	44.5		81.2% = $1 / 1.11^2$	36.1
2015	48.7		73.1% = $1 / 1.11^3$	35.6
2016		40.6 = $48.7 \times 5 / 6$	65.9% = $1 / 1.11^4$	26.7
2017		32.5 = $48.7 \times 4 / 6$	59.3% = $1 / 1.11^5$	19.3
2018		24.3 = $48.7 \times 3 / 6$	53.5% = $1 / 1.11^6$	13.0
2019		16.2 = $48.7 \times 2 / 6$	48.2% = $1 / 1.11^7$	7.8
2020		8.1 = $48.7 \times 1 / 6$	43.4% = $1 / 1.11^8$	3.5
BV _{1/1/2013}				1,000.0
Total Value				1,178.1

Solution 1

- b) Starting in 2013, this company entered into a new market other companies had not yet entered. The DDM method assumes dividend growth in perpetuity which may be unsustainable as competition enters the market. Whereas the AE method assumes abnormal earnings will decrease to zero over some forecasted time horizon which may be more realistic as competition enters the market.

Other acceptable answers may include but are not limited to:

The DDM method is highly leveraged to the terminal value and the assumptions that underlie the terminal value whereas the AE method is less leveraged to the terminal value.

The AE method focuses on the source of value creation – the company’s ability to earn a return on equity in excess of the investors’ required return. The DDM focuses only on the effect of this value creation – the company’s ability to pay cash flows to its owners.

The actual ROE is considerably higher than the required return, k , and these types of earning are unsustainable as competition enters the market.

Examiner Comment

For part a), the question specified that the abnormal earning would linearly decrease to zero by 2020, i.e., abnormal earning would be zero in the year 2020. However, a lot of candidates interpreted this as abnormal earnings decreasing to zero linearly by the year 2021. Both solutions, if calculated correctly, were accepted.

A common mistake was that candidates incorrectly identified the “expected equity market risk premium” as the “expected market return” therefore incorrectly calculating the required return, k . Another common mistake is that candidates did not include the company’s beginning book value in the calculated value of this company.

For part b), a mistake several candidates made was saying that the DDM would understate the value of the company. In this particular scenario, the DDM is actually higher than the AE valuation method. Additionally, some candidates lost at least partial credit for not fully assessing and explaining the reasoning behind their answer.